Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 30 minutes

🍌 Extract DNA from a Banana

You will learn: Soap breaks cell membranes, salt clumps DNA, and cold alcohol makes DNA come out of solution as white threads.

YOU NEED

- Half a ripe banana (or onion / strawberry)
- Zip bag
- Dish-washing liquid (1 spoon)
- Salt (half a spoon)
- Warm water (half a cup)
- Tea strainer or cloth
- Ice-cold rubbing alcohol (isopropyl, from a chemist) — keep in the freezer for 1 hour
- Clear glass, toothpick

⚠️ **Safety:** Rubbing alcohol is flammable and poisonous: keep it away from flames and never drink it. Do this with an adult.

STEPS

1. Mash the banana in the zip bag for 2 minutes.
2. Mix the soap, salt and warm water; add it to the bag and mash gently for 1 more minute (avoid foam).
3. Filter the mixture into the glass through the strainer.
4. Slowly pour ice-cold alcohol down the side of the glass to make a layer about as thick as the liquid.
5. Wait 2–3 minutes; white, stringy DNA appears between the layers.
6. Wind it on a toothpick and observe.

OBSERVATION TABLE

Step	Why we did it	Observation

QUESTIONS

1. Why did we add detergent?

2. Why does DNA rise into the alcohol layer?

3. Can you see a single DNA molecule this way? Why not?

WHAT IS HAPPENING? — THE SCIENCE

Detergent dissolves lipid membranes of the cell and nucleus, releasing DNA. Salt neutralises DNA's negative charges so strands clump. DNA is insoluble in cold alcohol, so it precipitates at the boundary — millions of DNA molecules together are visible.

Key terms: **DNA** **precipitation** **nuclear membrane**

Activity 2 · Hands-on · 🏫 In class · ⏱ 40 minutes

🧷 Paper Helix: Model Replication

> **You will learn:** DNA replication is semi-conservative — each new molecule has one old and one new strand.

YOU NEED

- Paper strips in 4 colours (A, T, G, C) and 2 colours for backbones (old = black, new = blue)
- Glue, scissors

STEPS

1. Make a 10-base-pair ladder: black backbones, bases paired A–T and G–C.
2. Twist it gently into a double helix.
3. "Unzip" it down the middle like helicase.
4. Build a new blue backbone with matching bases on each old strand.
5. Compare the two daughter molecules: how many black and blue strands does each have?

OBSERVATION TABLE

Original strand (5′→3′)	Complementary strand	New molecule 1 (old/new)	New molecule 2 (old/new)

QUESTIONS

1. How many hydrogen bonds join G and C?

2. Who proved semi-conservative replication, and how?

3. Why is the lagging strand made in pieces?

WHAT IS HAPPENING? — THE SCIENCE

Base pairing (A–T, G–C) means each strand can serve as a template. Meselson and Stahl proved with ^{15}N and ^{14}N that each daughter DNA keeps one parental strand — semi-conservative replication.

Key terms: **semi-conservative** **template** **complementary base pairing** **helicase**

Digital activity

💻 Gene to Protein Translator

> Type or pick a DNA template sequence and watch transcription and translation.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch5/2**

> **You will learn:** DNA template → mRNA (complementary, U instead of T) → codons read in threes from AUG → amino acids until a stop codon.

THINK & WRITE

1. What happens when a stop codon is reached?

2. Change one base in the middle. Did the protein change? (point mutation)